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Match Trends in Competitive Surgical Specialties Post ACGME Merger

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Abstract:	Background: Prior to the ACGME merger, residency programs were designated separately from MD and DO applicants. Competitive surgical specialties have historically been dominated by MD representation and spots for DO residents limited. However, the ACGME merger that took place in 2020 eliminated this separation. Objective: The aim of this study is to determine trends in DO representation in competitive surgical specialties by evaluating changes in overall DO representation and regional representation five years post-ACGME merger. Methods: Utilizing publicly accessible data, DO representation was evaluated across the following ACGME accredited competitive surgical specialties comparing PGY1 to PGY5: ENT, general surgery, orthopedic surgery, plastic surgery, and urology. Data was collected during the 2025 academic year. A total of 776 ACGME accredited residency programs were evaluated. Results: Based on the results, it was concluded that five years post ACGME merger, there was an increase in ENT and general surgery DO representation overall and a decrease in urology, orthopedic surgery, and plastic surgery. General surgery was found to have the most DO representation expansion with a 3% increase overall. Conclusion: In evaluating regional data, results indicate an increase in DO representation regionally in all specialties with the exception of orthopedic surgery and urology. In conclusion, there was minimal overall DO representation advancement across surgical specialties and regional distribution is varied.

Dear Editors,

On behalf of my co-authors, I am pleased to submit our manuscript entitled “Match Trends in Competitive Surgical Specialties Post ACGME Merger” for consideration for publication in the *Journal of Graduate Medical Education*. This manuscript has not been published previously and is not under consideration for publication elsewhere.

The study examines trends in DO representation in competitive surgical specialties by evaluating changes in overall DO representation and regional representation five years post-ACGME merger. We believe our findings are relevant to the *Journal of Graduate Medical Education* readership and contribute to advancing osteopathic medicine by addressing an important concern for osteopathic medical students and overall osteopathic medical education.

We confirm that all authors meet authorship criteria and have approved the final version of the manuscript. All relevant ethical standards have been upheld, including IRB approval. Conflict of interest and funding statements are included as required.

Thank you for considering our manuscript for publication in the *Journal of Graduate Medical Education*. We appreciate your time and look forward to your review.

Sincerely,

Ashley P Lesser, OMS IV, MBA, BS

On behalf of all authors

Title Page

Article Title

Match Trends in Competitive Surgical Specialties Post ACGME Merger

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Financial Disclosures : None reported.

Ethical Approval

IRB review is conducted by performing a risk-benefit analysis and applying HHS regulations at 45 CFR 46, which implement the National Research Act of 1974. The study was reviewed and approved through the Kansas City University International Review Board on August 19th, 2024. IRB approval number is 2226107-1.

Informed Consent

No informed consent needed.

Author Contributions

Ashley P. Lesser, Madison E. Kelly, Shay B. Van Sambeek and Christian J. Nassif provided substantial contributions to conception and design, acquisition of data, analysis and interpretation of data, and drafted the article. John P. Dobson gave final approval of the version of the article to be published; and *all authors* agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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ABSTRACT

Background: Prior to the ACGME merger, residency programs were designated separately from MD and DO applicants. Competitive surgical specialties have historically been dominated by MD representation and spots for DO residents limited. However, the ACGME merger that took place in 2020 eliminated this separation.

Objective: The aim of this study is to determine trends in DO representation in competitive surgical specialties by evaluating changes in overall DO representation and regional representation five years post-ACGME merger.

Methods: Utilizing publicly accessible data, DO representation was evaluated across the following ACGME accredited competitive surgical specialties comparing PGY1 to PGY5: ENT, general surgery, orthopedic surgery, plastic surgery, and urology. Data was collected during the 2025 academic year. A total of 776 ACGME accredited residency programs were evaluated.

Results: Based on the results, it was concluded that five years post ACGME merger, there was an increase in ENT and general surgery DO representation overall and a decrease in urology, orthopedic surgery, and plastic surgery. General surgery was found to have the most DO representation expansion with a 3% increase overall.

Conclusion: In evaluating regional data, results indicate an increase in DO representation regionally in all specialties with the exception of orthopedic surgery and urology. In conclusion, there was minimal overall DO representation advancement across surgical specialties and regional distribution is varied.

MAIN MANUSCRIPT: INTRODUCTION

Prior to the merger of the Accreditation Council for Graduate Medical Education (ACGME) and the American Osteopathic Association (AOA) accreditation systems that occurred in 2020, residency programs for allopathic (MD) and osteopathic (DO) applicants were designated separately. During a five-year transition period from 2015-2020, American Osteopathic Association (AOA) programs had to demonstrate compliance with ACGME requirements. The resulting single accreditation system streamlines graduate medical education by eliminating duplicative accreditation costs, ensuring uniform standards of evaluation and accountability, and expanding opportunities for all medical graduates—including access to osteopathic training for both allopathic and osteopathic students. It also fosters greater collaboration across the medical education community while providing transparency for residents.¹

Competitive surgical specialties have historically been dominated by MD representation with spots for DO residents limited. With the ACGME merge, this separation was eliminated, but the effects of this transition are not yet fully understood. There are limited studies that detail a comprehensive analysis across multiple surgical disciplines and geographic regions.

Prior studies reporting lower match rates for DO applicants compared to MD applicants in surgical fields. For example, a 2022 study reported that nearly half of surveyed surgical programs did not train DO residents,² and an additional study found lower DO match rates across multiple specialties.³ Similarly, a 2023 study demonstrated that DO applicants experienced significant losses in ophthalmology following the merger.⁴ Together, these findings reinforce the persistent underrepresentation of DOs in highly competitive surgical fields. However, this study adds a unique perspective to the current body of literature. Occurring five years after the ACGME merger this study is the most recent assessment of its effects. These findings help draw attention to the current under representation of DO students despite the ACGME merger. This highlights the need for further intervention to ensure equity between allopathic and osteopathic students.

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4 In this study, we aim to further evaluate representation of DO graduates across five ACGME-accredited competitive
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6 surgical specialties: ENT, general surgery, orthopedic surgery, plastic surgery, and urology. Understanding the
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8 distribution of DO physicians in these fields can provide valuable insight into the opportunities for osteopathic
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10 graduates to match into competitive surgical specialties compared to their MD counterparts. This data can be used to
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12 help future applicants, medical school advisors, and residency program directors better understand the trends that
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14 exist for their specialty of interest.
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18 **MAIN MANUSCRIPT: METHODS**

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22 The study protocol was reviewed and approved by the Kansas City University Institutional Review Board (approval
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24 number: 2226107-1). No funding was received to support this study.
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28 Residency program data were collected from the American Association of Medical Colleges (AAMC) website for
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30 otolaryngology,⁵ orthopedic surgery,⁶ and general surgery.⁷ Program listings for plastic surgery,⁸ and urology⁹ were
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32 obtained from the American Society of Plastic Surgeons and the American Urological Association, respectively.
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36 From the listings of programs, MD and DO degree data was obtained in the following manner: Each individual
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38 residency program's public website was accessed to retrieve resident demographic information including total
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40 resident numbers, total number of MDs and DOs, and numbers of MDs and DOs in PGY1 and PGY5. Excluded
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42 from the total number of residents were research fellows and those undergoing a program development year.
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45 Programs region was also noted based on the AAMC Residency Regions. This data was all collected between July
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47 of 2024 and December of 2024.
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51 The following inclusion parameters were applied to the initial listings of all residency programs across all specialties
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53 of interest (Table 1). The program was an US ACGME accredited program. The program had PGY1-PGY5
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55 residency classes. The program website contained resident and program director degree types or was publicly
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57 accessible. The program was a civilian residency program. Additionally, integrated and independent plastic surgery
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59 residency programs were separated, and only integrated programs were included. If any of these parameters were
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not met the program was excluded. This resulted in 117 ENT, 274 general surgery, 175 orthopedic surgery, 79 plastic surgery, and 131 urology residency programs evaluated in this study.

Microsoft Excel was used to analyze the data collected assessing for mean and percent change of both MD and DO populations in PGY1 and PGY5. A paired T-test was performed with regional data again comparing DO representation for PGY5 and PGY1.

MAIN MANUSCRIPT: RESULTS

ENT

DO residents account for 5.8% of all PGY5 ENT residents and 7.9% of PGY1 ENT residents. This depicts a 2.1% increase in DO residents from PGY5 to PGY1. DO representation across AAMC Residency Regions for PGY5 showed the East North Central region had the largest DO ENT resident population accounting for 16.7% while the West North Central, South Atlantic, Mountain, and Pacific regions had no DO ENT residents. For PGY1, the East North Central region had the largest DO ENT Resident population with 22.7% while the East South Central, Mountain, and Pacific Region had no DO ENT residents. Regional DO representation comparison between PGY5 and PGY1 is shown in Table 2. The East North Central region had the largest increase of 6.0% while the West South Central region demonstrated the largest decline of -25.1%. Statistical significance for regional change utilizing Paired T-test indicated no statistical significance for regional changes ($p=0.57$).

General Surgery

General Surgery DO residents account for 13.9% of all PGY5 residents and 17.1% of PGY1 general surgery residents. This depicts a 3.2% increase in DO residents from PGY5 to PGY1. DO representation comparison across AAMC Residency Regions for PGY5 to PGY1 is depicted in Table 2. For PGY5, the East North Central region had the largest DO ENT resident population accounting for 26.7% and again the New England region had the least with 1.1%. For PGY1, the East North Central region had the largest DO general surgery resident population with 24.3% while the New England region had the least with 5.3%. All regions saw an increase in DO resident representation five years following the ACGME merger with the exception of the East North Central Region showed a decrease of

-2.5%. The South Atlantic region had the largest increase of 9.9%. Statistical significance for regional change utilizing Paired T-test indicated statistical significance for regional changes ($p=0.005$).

Orthopedic Surgery

DO residents account for 12.7% of all PGY5 orthopedic surgery residents and 12% of PGY1 residents. DO regional representation for PGY5 demonstrated the similar distribution with the East North Central having the highest representation and the New England region again having the lowest. For PGY1, the East North Central region had the largest DO orthopedic surgery resident population with 27.2% while the New England region had the least with 2.0%. DO representation comparison across AAMC Residency Regions for PGY5 to PGY1 is depicted in Table 2. Statistical significance for regional change utilizing Paired T-test indicated no statistical significance for regional changes ($p=0.31$).

Plastic Surgery

DO plastic surgery residents account for 2 % of all PGY5 residents and 1.1% of PGY1 residents. DO representation across AAMC Residency Regions for PGY1 and PGY5 plastic surgery DO demographics are depicted in Table 2. DO regional representation for PGY5 demonstrated DO residents only in the East North Central and South Atlantic regions. For PGY1, the East North Central region and the Middle Atlantic region were the only regions with DO plastic surgery residents. Statistical significance for regional change utilizing Paired T-test indicated no statistical significance for regional changes ($p=0.30$).

Urology

DO urology residents account for 10% of all PGY5 orthopedic surgery residents and 8.5% of PGY1 residents. There was a 1.5% decrease in DO representation from PGY5 to PGY1. DO representation across AAMC Residency Regions for PGY1 and PGY5 urology surgery DO demographics are depicted in Table 2. DO regional representation for PGY5 demonstrated the highest DO representation in the Middle Atlantic region and no DO residents in the New England, West North Central, Pacific, and Mountain regions. For PGY1, the South Atlantic region had the largest DO urology resident population with 10.3% while the New England region, East South

Central, and Pacific had no DO residents. Statistical significance for regional change utilizing Paired T-test indicated no statistical significance for regional changes ($p=0.39$).

MAIN MANUSCRIPT: DISCUSSION

Overall, the findings suggest limited benefits of the ACGME merger for osteopathic applicants in competitive surgical specialties. Although DO representation increased in general surgery and otolaryngology, decreases in orthopedic surgery, plastic surgery, and urology underscore persistent differences. These results challenge the ACGME's assertion that the merger "enhanced opportunities" for DO applicants, indicating that systemic barriers may remain.

The results of this study found that there was an increase in overall representation for DO applicants in ENT and General Surgery. However, there was a decrease in overall representation in Orthopedic Surgery, Plastic Surgery, and Urology. These overall trends are evident in Figure 1. Regional changes were found to not be statistically significant in ENT, Orthopedic Surgery, Plastic Surgery, and Urology. However, General Surgery had an increase in DO representation in every region except the East North Central which showed a decrease. The largest representation for General Surgery was in the East North Central region and the least in the New England region for both PGY1 and PGY5 residents. Overall regional trends were also identified, with overall regional DO distribution remaining highest in the East North Central region and lowest in the New England Region, no change was noted in the Pacific and Mountain regions, however redistribution was noted in the West North Central, West South Central, East South Central, and South Atlantic regions as appreciated in Figure 2.

Although there was the largest increase in representation in General Surgery overall and regionally, the specialty itself is the least competitive of those analyzed. The increase in representation in the ENT group is far outweighed by the decrease in representation in the Orthopedic Surgery, Plastic Surgery, and Urology groups.

Prior research has consistently shown lower match rates and underrepresentation of DO applicants in competitive surgical specialties, with some studies noting significant post-merger declines. A study published in 2025 compared

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4 match rates of DO students before and after the ACGME merger. This study aligned with our results showing a
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6 significant increase in DO match rates in general surgery, but no change in integrated residency positions like plastic
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8 surgery and thoracic surgery. Our study, conducted five years after the ACGME merger, provides the most recent
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10 assessment and underscores the ongoing need for interventions to promote equity between MD and DO applicants.
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14 While this study expands upon the current literature it is important to acknowledge certain limitations. One such
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16 limitation is the collection of data from only five surgical specialties. There are other competitive surgical specialties
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18 such as Neurosurgery, Vascular Surgery, Obstetrics and Gynecology, Cardiothoracic Surgery, and Ophthalmology
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20 which may have provided more insight to our study. Another constraint was the limited access to resident
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22 information for certain programs. These programs were excluded from the study but may have increased the
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24 statistical power. Finally, since there was an increase in representation among ENT and General Surgery the data
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26 may be interpreted differently and as increasing representation for DOs.
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31 **MAIN MANUSCRIPT: CONCLUSIONS**

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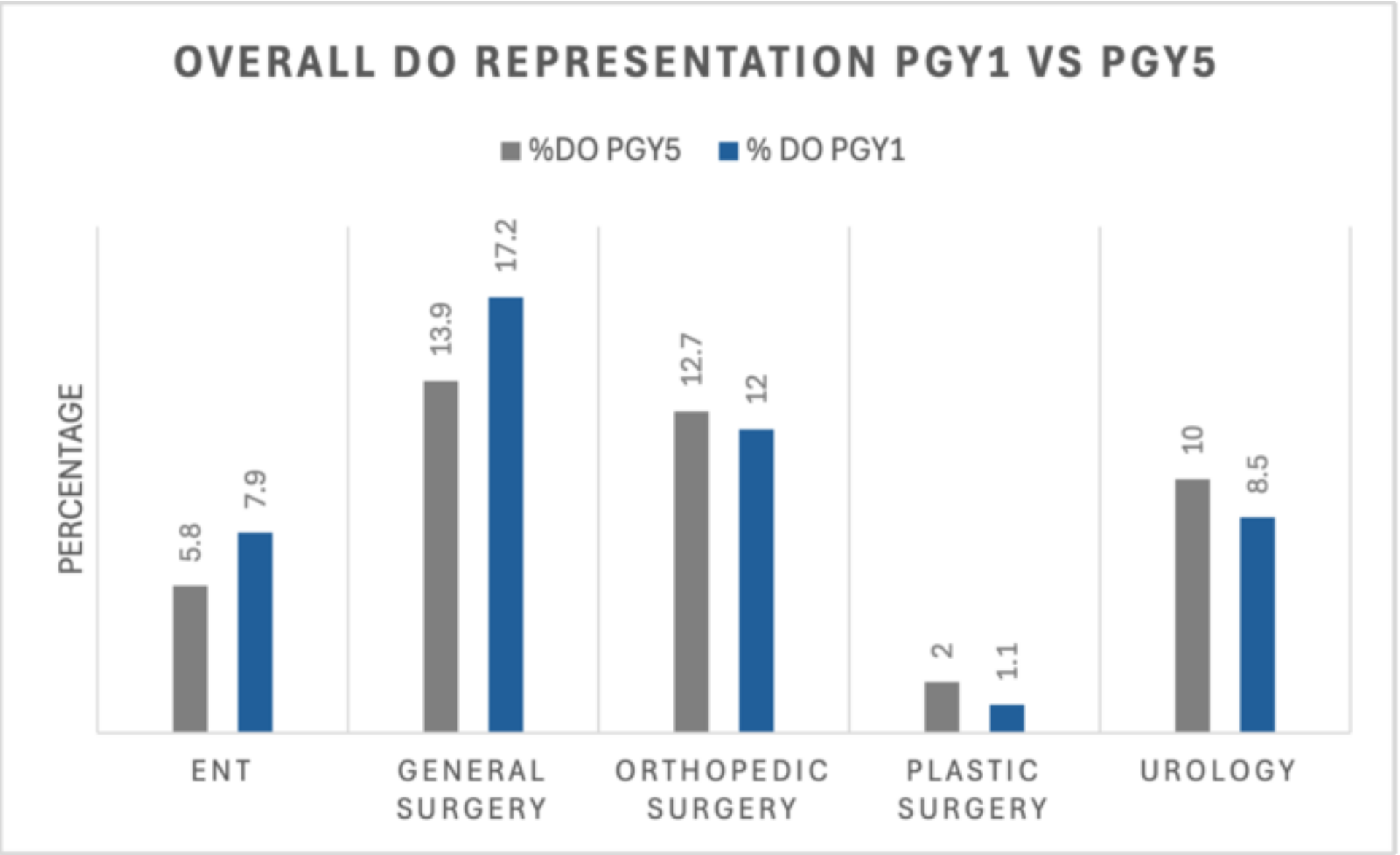
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34 Following the ACGME merger completed five years ago, osteopathic medical students have seen
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36 minimal improvement in representation throughout the exploration of the five surgical specialties
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38 detailed in our investigation with significant variation in regional distribution. Although increases
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40 were observed in otolaryngology and general surgery, decreases in orthopedic surgery, plastic surgery,
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42 and urology highlight ongoing differences. Regional analysis revealed heterogeneous trends, with
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44 some areas showing modest growth while others demonstrated declines or stagnation. Further research
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46 should evaluate additional competitive specialties, program-level factors such as leadership
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48 composition, and applicant characteristics associated with successful DO matches. Our hope is that as
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50 the number of DO applicants increase, we see a linear relationship in this representation in
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52 competitive surgical specialties.
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Figure 1 _ Overall DO Representation PGY1 vs PGY5



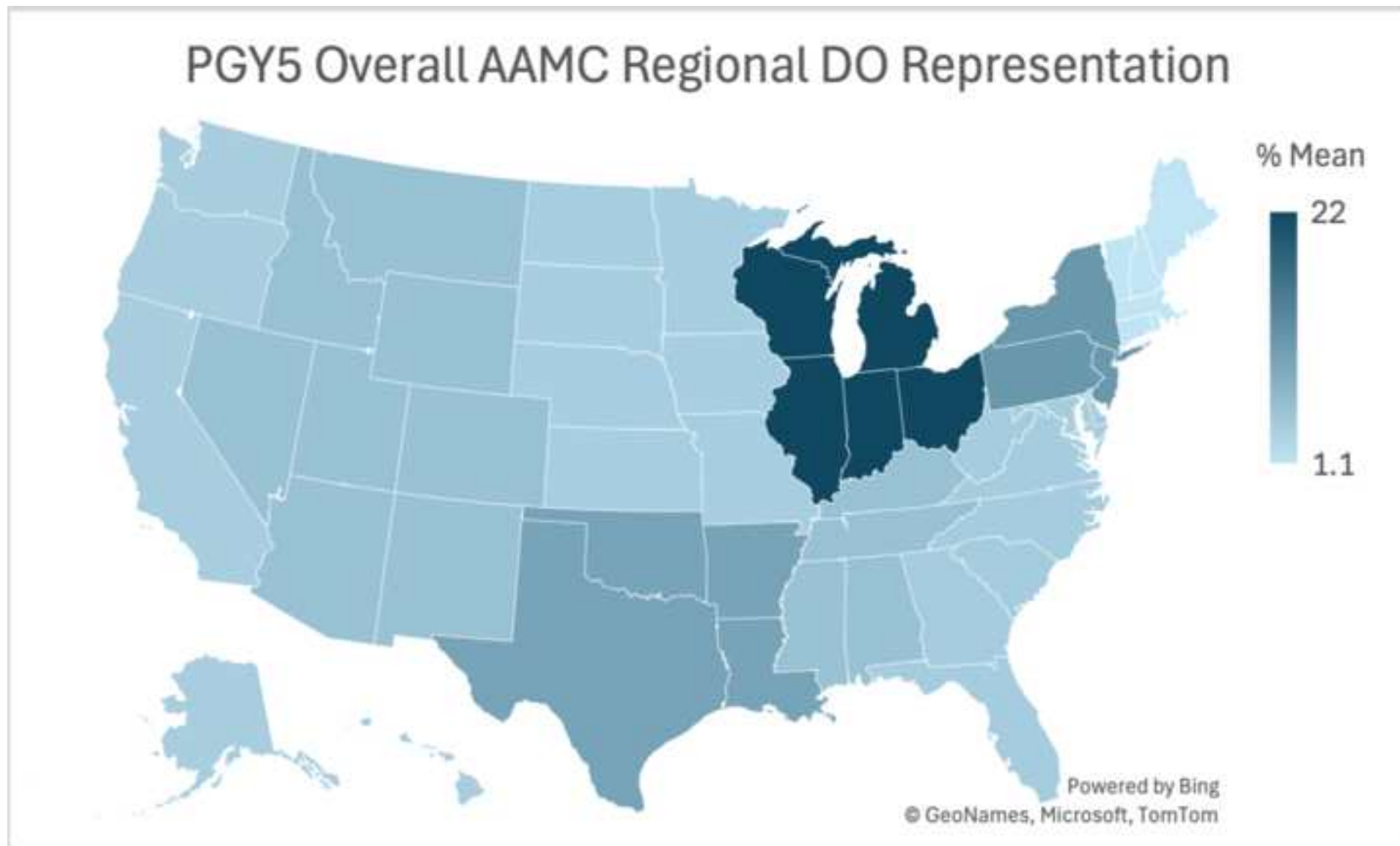




Table 1

Program Inclusion Parameters	
#	Inclusion Parameter
1	Program is a United States ACGME accredited program.
2	Program includes residency classes PGY1 (Class of 2029) through PGY5 (Class of 2025).
3	Program website contains resident degree type or degree was publicly accessible (MD or DO).
4	Program is a civilian residency program.
5*	Program is Integrated.

*Denotes Parameter Specific to Plastic Surgery Residency Programs

Table 2

Comparison of MD and DO representation across AAMC Residency Region for ENT, General Surgery, Orthopedic Surgery, Plastic Surgery, and Urology for classes PGY5 and PGY1.

Specialty	Region	PGY5 %MD	PGY5 %DO	PGY1 %MD	PGY1 %DO
ENT	New England	95.7	4.3	95.2	4.8
ENT	Middle Atlantic	88.1	11.9	88.3	11.7
ENT	East North Central	83.3	16.7	77.3	22.7
ENT	West North Central	100	0	94.6	5.4
ENT	South Atlantic	100	0	98.3	1.7
ENT	East South Central	95.2	4.8	100	0
ENT	West South Central	70	30	95.1	4.9
ENT	Mountain	100	0	100	0
ENT	Pacific	100	0	100	0
General Surgery	New England	98.9	1.1	94.7	5.3
General Surgery	Middle Atlantic	82.1	17.9	79.2	20.8
General Surgery	East North Central	73.2	26.8	75.7	24.3
General Surgery	West North Central	81.4	18.6	76.5	23.5
General Surgery	South Atlantic	93.8	6.2	83.9	16.1
General Surgery	East South Central	91	9	83.5	16.5
General Surgery	West South Central	93.3	6.7	91	9
General Surgery	Mountain	83.9	16.1	79.3	20.7
General Surgery	Pacific	93.8	6.2	91.4	8.6
Orthopedic Surgery	New England	100	0	98	2
Orthopedic Surgery	Middle Atlantic	82.1	17.9	85.5	14.5
Orthopedic Surgery	East North Central	76.4	23.6	72.8	27.2

Orthopedic Surgery	West North Central	98.6	15.4	95.5	4.5
Orthopedic Surgery	South Atlantic	93.1	6.9	94.6	5.4
Orthopedic Surgery	East South Central	89.5	10.5	92.1	7.9
Orthopedic Surgery	West South Central	96.1	3.9	96.3	3.7
Orthopedic Surgery	Mountain	87.9	12.1	96.9	3.1
Orthopedic Surgery	Pacific	85.6	14.4	89.1	10.9
Plastic Surgery	New England	100	0	100	0
Plastic Surgery	Middle Atlantic	100	0	96.2	3.8
Plastic Surgery	East North Central	93.1	6.9	96.9	3.1
Plastic Surgery	West North Central	100	0	100	0
Plastic Surgery	South Atlantic	95.7	4.3	100	0
Plastic Surgery	East South Central	100	0	100	0
Plastic Surgery	West South Central	100	0	100	0
Plastic Surgery	Mountain	100	0	100	0
Plastic Surgery	Pacific	100	0	100	0
Urology	New England	100	0	100	0
Urology	Middle Atlantic	90.9	9.1	91.8	8.2
Urology	East North Central	63.8	36.2	80	20
Urology	West North Central	100	0	95.7	4.3
Urology	South Atlantic	96.2	3.8	97	3
Urology	East South Central	95.2	4.8	100	0
Urology	West South Central	92.1	7.9	89.7	10.3
Urology	Mountain	100	0	92.3	7.7
Urology	Pacific	100	0	100	0